



**TRƯỜNG ĐẠI HỌC
CÔNG NGHỆ ĐÔNG Á**
EAST ASIA UNIVERSITY OF TECHNOLOGY

International Training & Cooperation Institute

Introduction to Blockchain Technology

- ❖ Lesson 0 - Lecture Overview
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- ❖ Module Blockchain Technology is a module of the specialized knowledge of Information Systems.
- ❖ This module can help students understand the principles of Blockchain Knowledge; How to create Block; mechanism of forward movement in Block. Grasp the knowledge to build a Smart Contract based on the Blockchain platform. Create wallets, wallet addresses and transfer coins to wallets together.
- ❖ After finishing this module, students can adapt and install, and configure to create their own Token.

Text Book:

We'll be following the curriculum in the textbook:

❖ Blockchain Technology: Exploring Opportunities, Challenges, and Applications, 1st Edition, 2022

I will also update the content from other sources to catch up with the development of Blockchain Technology



Grading Structure



❖ 10% Attendance, Quizzes or Homework

❖ 30% Mid-term

❖ 60% Final exam or Final Project

Quizzes or Homework



- ❖ We will have small quizzes or homework to reinforce material and let you know what you need to study.
- ❖ When we have an in-class quizzes, I will announce them at least a week ahead of time in class. I will also let you know in advance if a homework is used for evaluation.

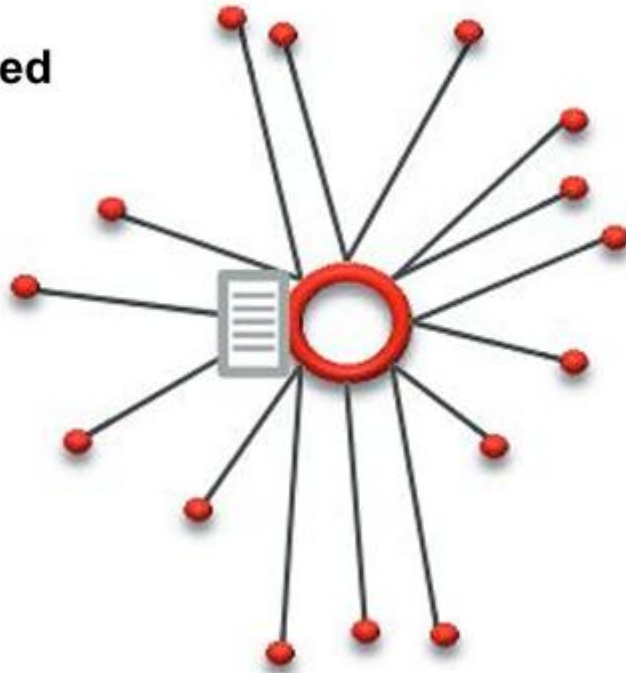
❖ We'll have two exams:

- Mid Term: This will definitely cover the early sections. It probably a writing exam
- Final Exam: This will be comprehensive. We'll focus mostly on the newer material. It can be a writing exam or final project.

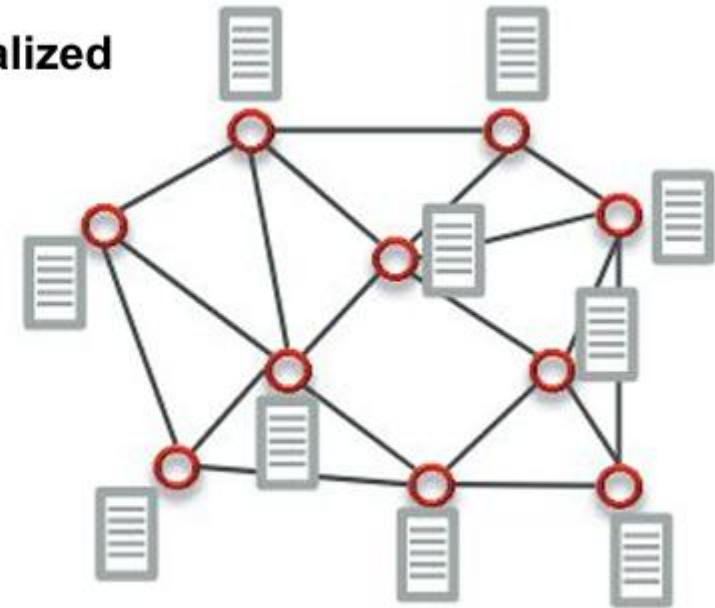
A Quick Look at Blockchain

- ❖ Blockchain is a decentralized distributed ledger technology that runs over a peer-to-peer (P2P) network

Centralized



Decentralized



Who invented it?



❖ Bitcoin

- Digital cryptocurrency
- Authored in white paper in 2008 by Satoshi Nakamoto and launched in 2009
- First blockchain database to keep transactions records

❖ Satoshi Nakamoto

- Pseudonymous person or persons
- Maybe an organization
- Two years later (April 2011) disappears from the web

Satoshi Nakamoto - Who is he?

❖ Helger Lipmaa?



❖ Craig Wright?



❖ Elon Musk?



Blockchain before Bitcoin



Bitcoin is built on earlier work:

- ❖ Peer-to-peer/distributed computing
- ❖ Cryptography and Merkle Tree
- ❖ Cryptographically linking blocks
 - Stuart Haber and W. Scott invented in 1991
- ❖ Smart contracts
 - Nick Szabo in 1997 introduced the idea of smart contracts
- ❖ Digital currency
 - Nick Szabo in 2005 introduced **Bitgold** as a form of digital money



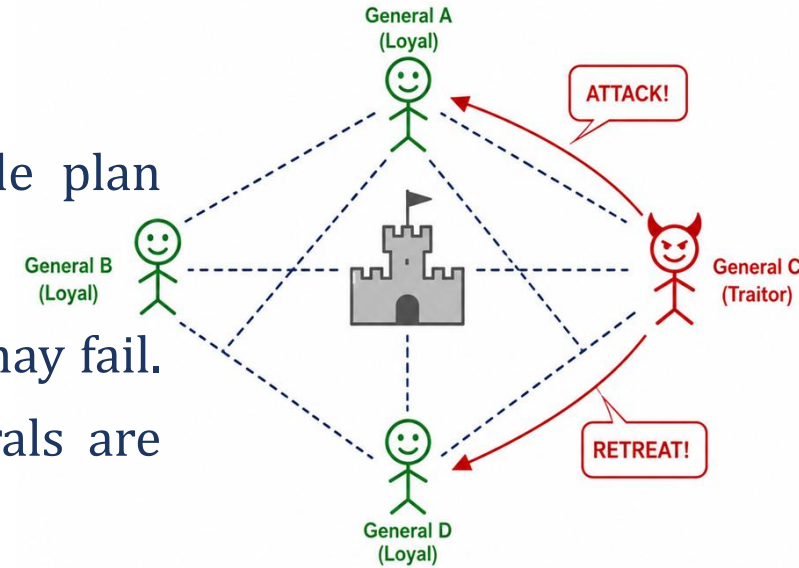
What made Bitcoin different with the Previous Works?



- ❖ Decentralized distributed **consensus mechanism** and **incentive layer**
 - Proof of work
 - Crypto mining
 - Crypto economics
- ❖ Satoshi put together the already available technology along with consensus mechanism and introduced an incentive layer
 - It overcomes the **centralization** and **double-spending issues** in digital currency

Theoretical Problem - The Byzantine Generals Problem (BGP)

- ❖ Proposed by Lamport, Shostak, and Pease in 1982.
- ❖ **Scenario:** A group of generals must agree on a common battle plan (Attack or Retreat). They can only communicate via messengers.
- ❖ **Problem:** Some generals (nodes) may be traitors or messengers may fail. How do they reach a **consensus** if more than $1/3$ of the generals are malicious?



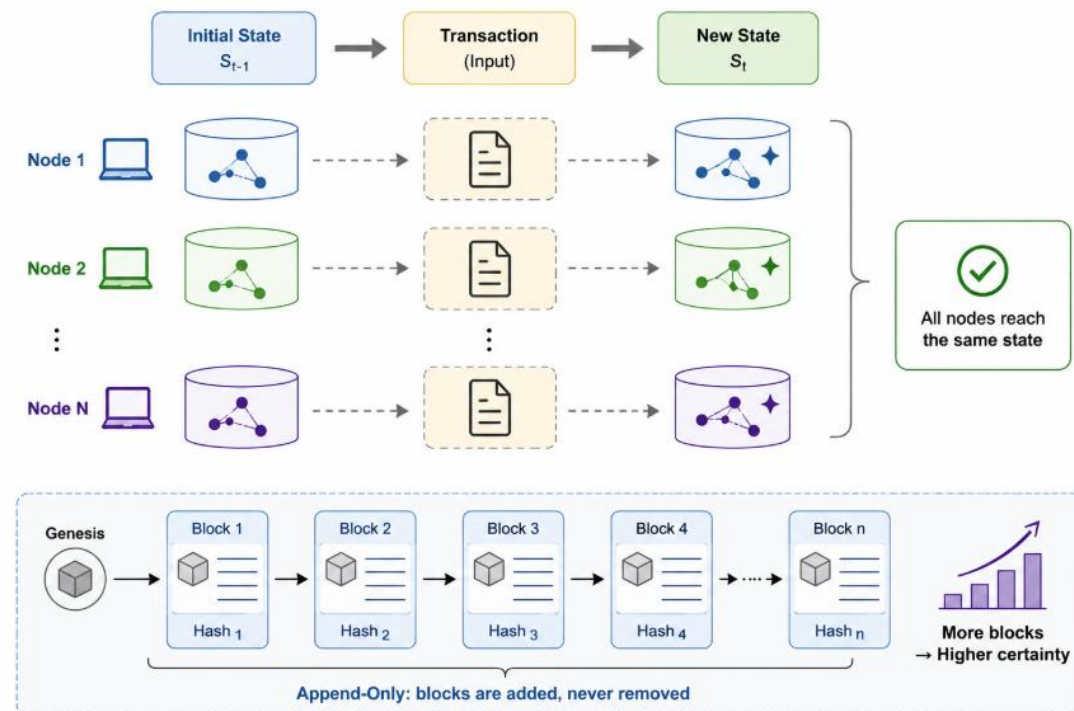
- ❖ After comparing messages:
 - A sees that C sends inconsistent messages.
 - B sees the same.
 - D also realizes C is dishonest.
 - The loyal generals trust each other and choose **ATTACK** (3 loyal votes).

General	Sends
A (Loyal)	ATTACK
B (Loyal)	ATTACK
C (Traitor)	ATTACK to A, RETREAT to D
D (Loyal)	ATTACK

=> **Consensus = ATTACK**

Blockchain: A Probabilistic State Machine

- ❖ **Technical Definition:** Beyond a **distributed ledger**, blockchain is a Probabilistic State Machine
- ❖ **Deterministic Execution:** Every node executes the same input to reach the same system state
- ❖ **Probabilistic Finality:** Unlike traditional databases with instant finality, blockchain certainty increases as more blocks are appended (especially in PoW systems)
- ❖ **Append-Only Structure:** A sequential flat-file list of transaction records where entries are never deleted



Blockchain in the 4th Industrial Revolution



- ❖ Blockchain follows the evolution of Mainframes → PCs → Internet → Mobile/Social
- ❖ The World Economic Forum estimates that \$1 invested in digital technology (like blockchain) can increase GDP by \$20
- ❖ Moving from the "Internet of Information" to the "Internet of Value" where assets are transferred without intermediaries

Example: *Comparison of traditional bank wire transfers (3–5 days, multiple intermediaries) vs. Blockchain value transfer (near real-time, peer-to-peer)*

Blockchain Myths vs. Realities



- ❖ Myth: Blockchain is a database that lives in the cloud.
- ❖ Reality: Blockchain is an application that nodes download and run on their own computers. It is a tool used to prove something exists, not a storage server for files.
- ❖ Myth: You can store all your PDFs and large files directly on the blockchain.
- ❖ Reality: It only stores "Proof-of-Existence" using small digital codes called hashes. The actual files stay in separate storage systems called "Data Lakes" off-chain.
- ❖ Myth: Blockchain is 100% impossible to change.
- ❖ Reality: It is very hard to change because it is an "append-only" structure (you only add data, never delete). However, if 51% of the network participants agree, they can modify the data.

Example: *To protect a land record, you store the hash of the deed on the blockchain so no one can fake it. The large photo of the property is kept on a private company server.*

Q1: Should blockchain be viewed as a software protocol or a database?

Q2: If blockchain is **Trustless**, does it mean we don't need trust at all?

Q3: If a private company already has a secure SQL database, what specific technical advantages does a blockchain provide?